

What Determines How Much You Pay for an Apartment in the United States?

Course: Applied Statistics and Data Analysis (ASDA) – Leuphana University, WiSe 25/26

Interactive Dashboard: [Link](#)

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1. Introduction

Imagine you are searching for an apartment in the United States. You open a rental website and are immediately overwhelmed: a studio in Boston for \$2100, a two-bedroom in Ohio for \$650, a one-bedroom in San Francisco with a pool for \$2800. Prices seem to follow no logic. Or do they?

This report sets out to answer a deceptively simple question: **what drives apartment rent prices, and can we predict them?** We investigate a dataset of 10000 rental listings scraped from US property websites, covering 51 states, 1572 cities, and a rich mix of apartment sizes, amenities, and locations. Along the way, we uncover a clear hierarchy of price drivers and a few surprises.

Item	Description
Dataset name	apartments_for_rent_classified_10K
Time period	September–December 2019 (~3.3 months)
Number of rows	10 000 (raw)
Number of columns	22 (raw dataset)
Format	.csv (semicolon-delimited, cp1252 encoding)
Source	apartments for rent classified 10K.csv

Each listing contains the apartment's price, physical attributes (square footage, bedrooms, bathrooms), geographic information (city, state, GPS coordinates), a free-text **amenities description, pet policy, and listing metadata**. The key variables are summarised below:

Feature	Type	Description	Example values
price	int64	Listed rent	1200, 850, 2400
square_feet	int64	Area in square feet	750, 1100
bedrooms	float64	Number of bedrooms	0, 1, 2, 3
bathrooms	float64	Number of bathrooms	1, 1.5, 2

amenities	object	Comma-separated amenity list	"parking, pool, laundry"
pets_allowed	object	Pet policy	Cats, Dogs, None
state	object	US state abbreviation	CA, TX, NY
cityname	object	City name	Boston, Chicago
latitude	float64	GPS latitude	42.36, 34.05
longitude	float64	GPS longitude	-71.06, -118.24

2. Cleaning the Data

Web-scraped data is inherently messy, and this dataset was no exception. Before asking any analytical questions, we had to clean up several issues.

Near-constant columns. Our first check was whether any columns carry almost no information – where a single value dominates over 90% of entries. The result was striking:

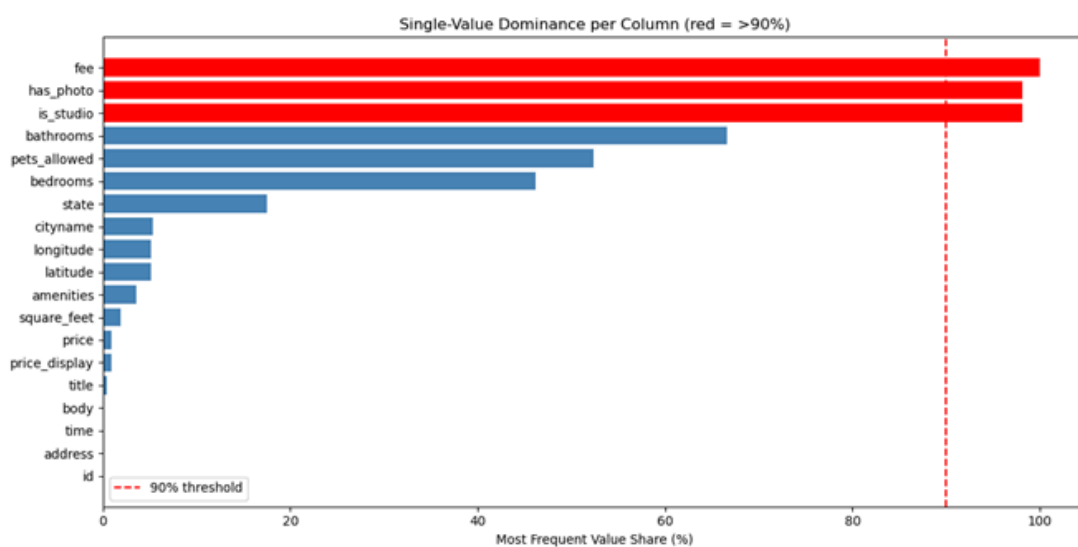


Fig 1: Single-value dominance per column – columns crossing the red 90 % threshold carry almost no information.

Three columns crossed the threshold: **fee** (100% identical), **has_photo** (98.2% "Thumbnail"), and **is_studio** (98.1 % "No"). These, along with **source**, **currency**, and **category**, were dropped.

Other cleaning steps are summarised below:

Issue	Columns affected	Action taken
Missing values	bathrooms (34), bedrooms (7)	Dropped rows with null values.
Invalid entries	bathrooms	Removed rows with 0 bathrooms (data-entry errors).
Mixed price periods	price, price_type	Converted all prices to monthly (weekly × 4, daily × 30). Dropped price_type .
Missing categorical values	pets_allowed (42 % null)	Filled NaN with "Not specified" to retain rows.

Outlier removal. Perhaps the most impactful step. Before cleaning, boxplots told a dramatic story – some listings claimed rents of \$52500 a month and apartments of 11000 sq ft. These are clearly erroneous or ultra-luxury outliers that would distort every model we build so we applied the **IQR method**, removing 615 price outliers (bounds: \$0–\$2814) and 511 square-feet outliers (bounds: 12–1688 sq ft).

3. A First Look at the Market

With clean data in hand, we asked the most basic question: **what does a "typical" apartment look like?**

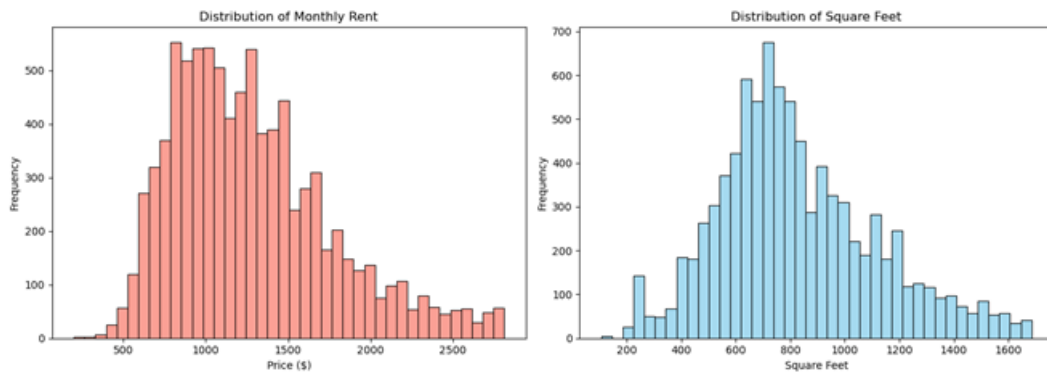


Fig 2: Distribution of monthly rent (right-skewed, mean \$1 281, median \$1 199) and square footage.

The typical apartment rents for about **\$1200/month** (median) and covers **~850 sq ft.** This skewness is important: it tells us that a log-transformation will be necessary later when we build regression models, because linear models assume normally distributed residuals.

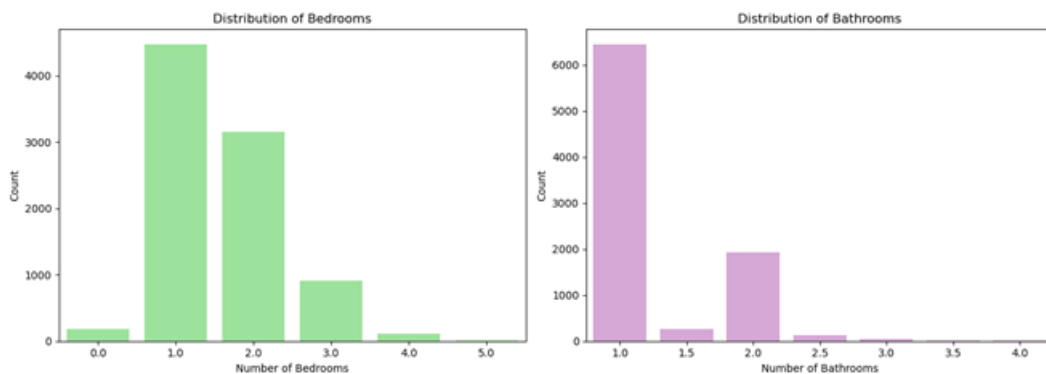


Fig 3: Bedrooms and bathrooms: 1-bed units dominate (46 %), most apartments have 1 bath (67 %).

The market is dominated by **1-bedroom apartments (46%)** and **1-bathroom units (67%)**. Studios make up only about **2%** of listings, and anything above 4 bedrooms is exceedingly rare.

So far, the picture is clear: a typical US rental is a 1-bed/1-bath apartment of about **850 sq ft** for **\$1200** a month. But averages hide the interesting story. The *variance* in rent is enormous – and that is what we set out to explain.

4. What Drives Rent?

4.1 The Obvious Suspect: Size

The first hypothesis is intuitive: bigger apartments should cost more. A scatterplot confirms this:

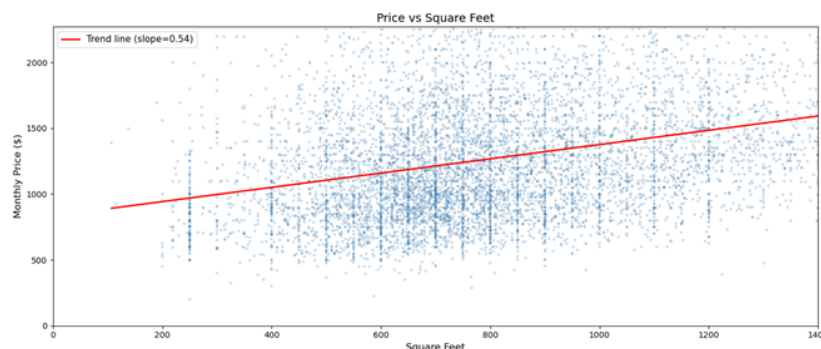


Fig 4: Price vs Square Feet: a clear upward trend, but enormous scatter.

The trend line slopes upward, confirming a positive relationship. But look at the scatter – at 800 sq ft, rents range from under \$500 to over \$2500. Size matters, but it is far from the whole story.

A hexbin density heatmap reveals where the market is thickest:

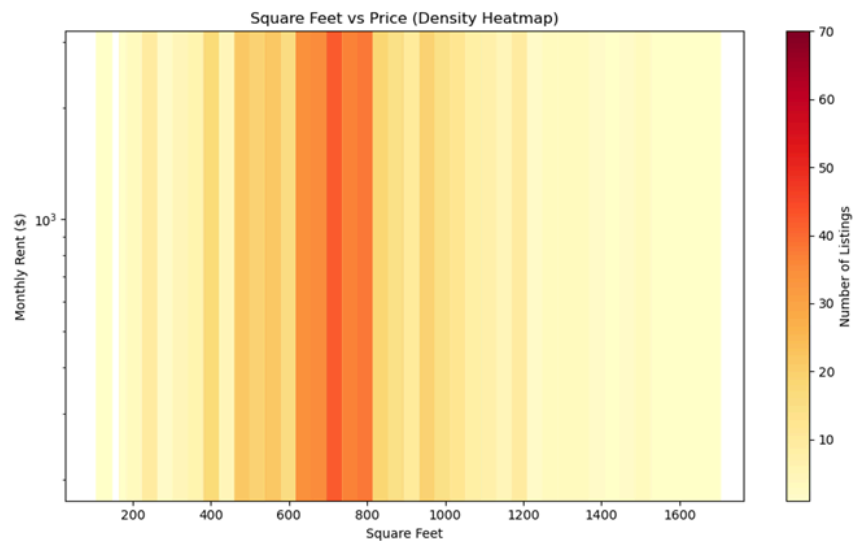


Fig 5: Hexbin density: the "sweet spot" of the market sits between 600–800 sq ft. The vertical spread at every size confirms that other forces are at play.

Most inventory clusters around 600–800 sq ft, exactly where the typical 1- and 2-bedroom units land. The wide vertical spread within this band is our first clue that **something beyond size is driving prices**.

4.2 The Value Paradox: Price per Square Foot

If bigger is better, do you at least get a better *deal* as apartments grow larger? We explored this with a flexible, non-linear trend that follows the data rather than forcing a straight line:

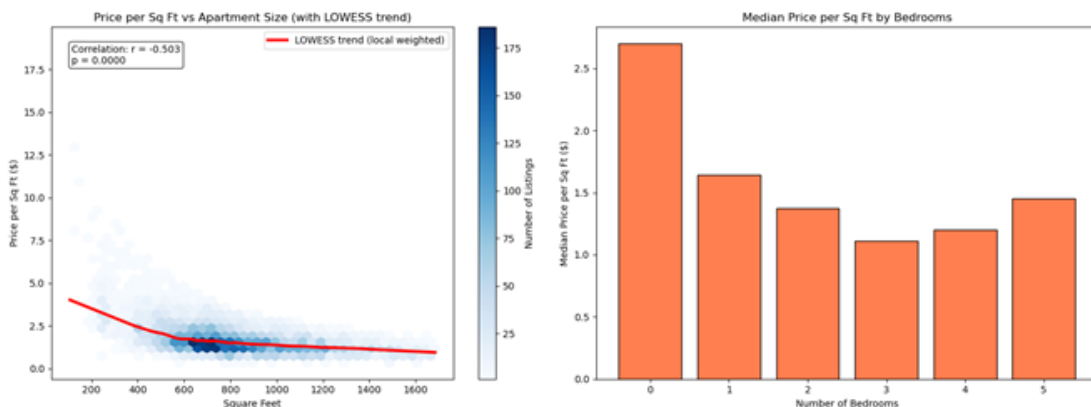


Fig 6: LOWESS analysis: price per sq ft plummets for larger units. Studios are the worst deal per foot; 3-bed units are the best.

The answer is a resounding **yes**. Price per square foot starts very high for tiny studios and drops sharply as size increases, leveling off around 1000 sq ft. The bar chart tells the same story by bedroom count: studios are the most expensive per foot (\$2+ per sq ft), while 3-bedroom units offer the best value. The slight uptick at 5 bedrooms reflects a luxury premium on rare, oversized units.

This is the "bulk value" effect: renters pay a premium for small spaces, likely because studios are concentrated in expensive city centres where space itself is scarce.

4.3 The Bedroom Puzzle

We next looked at how total rent varies by bedroom count. The result contained a surprise:

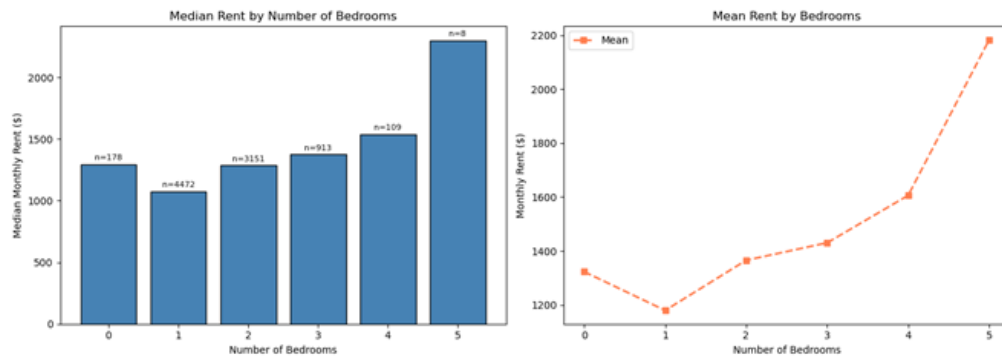


Fig 7: Median rent by bedrooms: studios cost *more* than 1-bedroom apartments – an urban-core effect.

Studios cost more on average than 1-bedroom apartments. This seems paradoxical: **why would less space cost more?** The explanation is geographic: studios are disproportionately located in expensive urban cores (**Manhattan, San Francisco, Boston**) where even the smallest apartment commands a premium. Once past the 1-bedroom mark, rent climbs steadily with each additional bedroom, as expected.

But are these differences statistically significant, or could they be noise? Before reaching for ANOVA, we checked the two key assumptions – normality and equal variance:

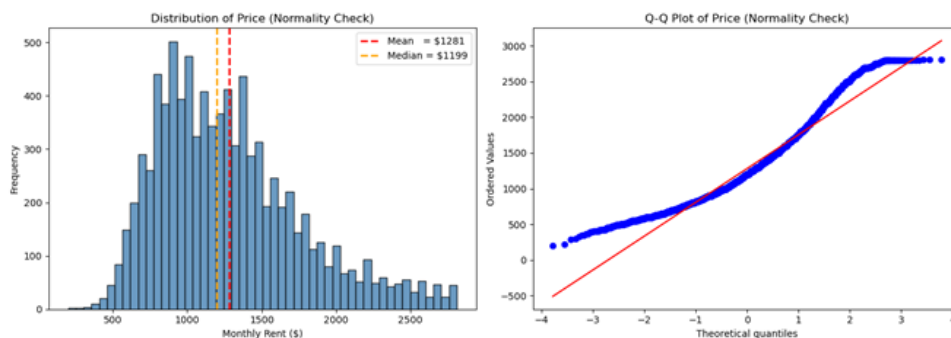


Fig 8: The price distribution is right-skewed, departing from normality.

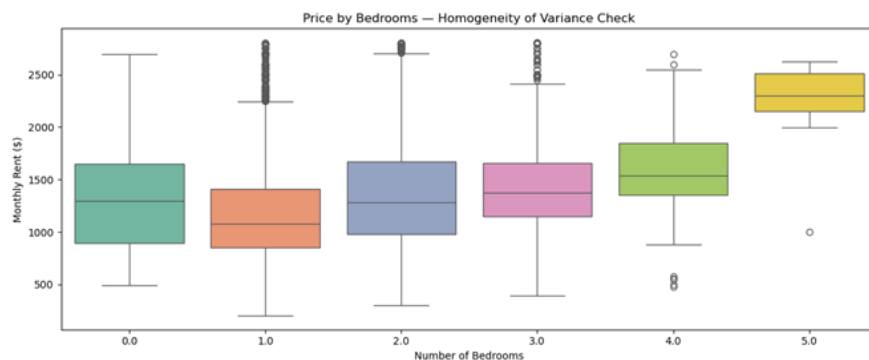


Fig 9: Variance increases with bedroom count – the equal-variance assumption is violated.

The Q-Q plot shows departure from normality in the tails, and the boxplot reveals clearly unequal variances across groups. Standard ANOVA would be unreliable here. We therefore used **Welch's ANOVA**, which is robust to unequal variances:

Welch's ANOVA: $F(5, 73.7) = 93.88$, $p < 0.001$, $\eta^2 = 0.051$

The result is highly significant – rents genuinely differ across bedroom groups, not by chance. A **Games-Howell post-hoc test** confirmed that nearly all pairwise comparisons are significant, with two

notable exceptions: studios vs. 2-bed and studios vs. 3-bed, where the difference does not reach significance – further evidence of the studio "urban premium" distortion.

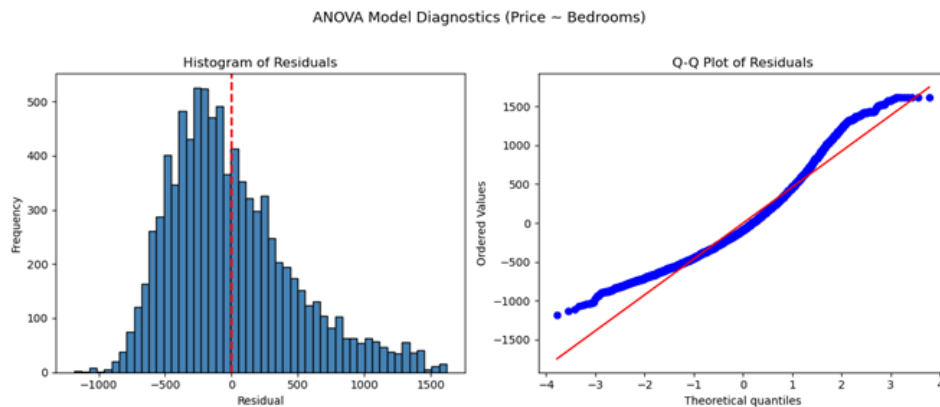


Fig 10: ANOVA residual diagnostics: the residuals show the right-skewed pattern, reinforcing why Welch's variant was necessary.

4.4 The Geography Factor: Where You Live Matters Most

Size and bedrooms tell part of the story. But our scatterplot(Fig 4) showed apartments of the same size differing by over \$1000. The missing variable? **Location.**

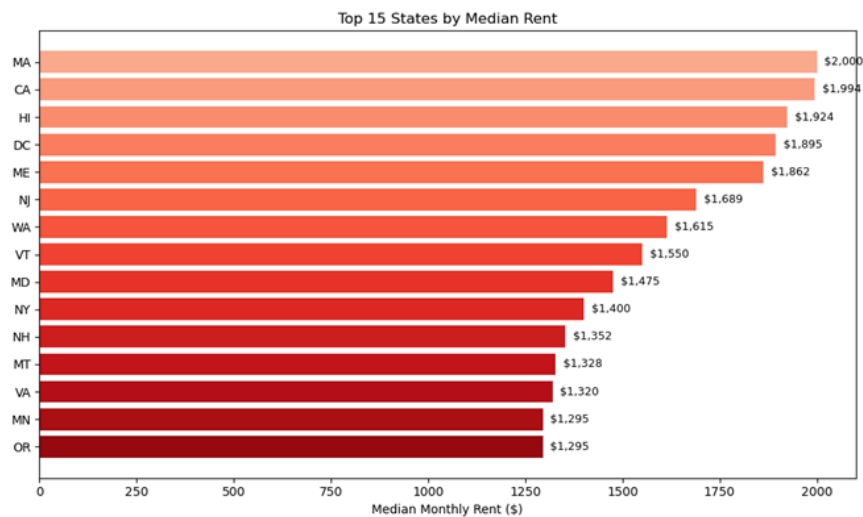


Fig 11: Top 15 states by median rent: Massachusetts (\$2000) and California (\$1994) tower over the rest.

The state-level picture is dramatic. Massachusetts leads with a \$2000 median rent, followed closely by California at \$1 994. At the bottom, many midwestern and southern states sit well under \$1000. That is a **2x price difference** for equivalent apartments, driven entirely by geography.

But states are large. **Which cities are actually doing the heavy lifting?**

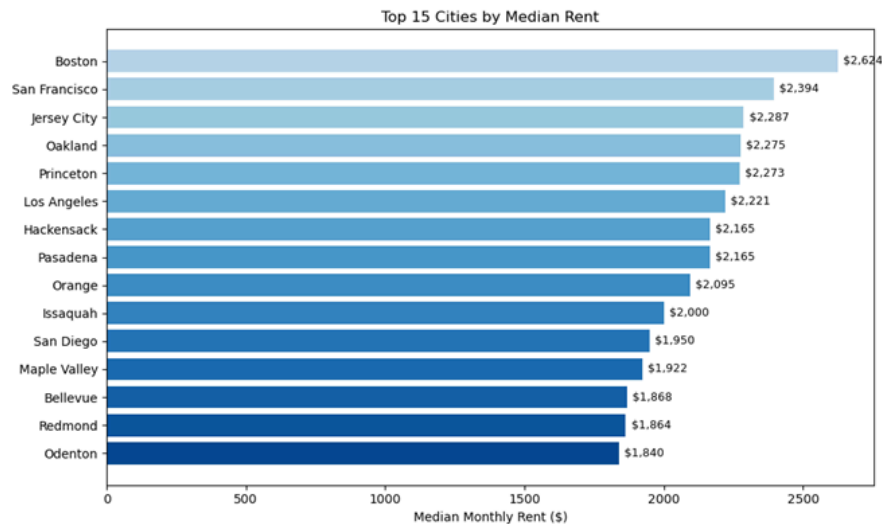


Fig 12: Top 15 cities by median rent: Boston, San Francisco, and San Diego are the big three.

The answer is clear. Boston's extreme rents single-handedly push Massachusetts to the #1 spot. San Francisco, Los Angeles, and San Diego do the same for California. These "expensive" states are not uniformly pricey – they are pulled up by a handful of high-demand cities that charge a massive premium. **The most important decision a renter makes is not how many bedrooms to get – it is which city to live in.**

4.5 A Temporal Footnote

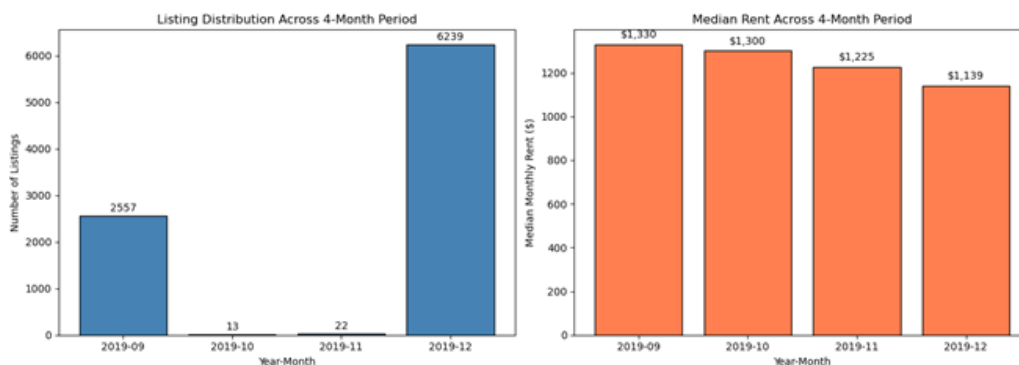


Fig 13: The dataset spans only ~3.3 months – far too short for seasonality analysis.

Before moving on, we checked whether rent varied over time. The dataset covers only September–December 2019 (~3.3 months). Rents appear stable across this window, but the time span is far too short to draw any conclusions about seasonality. This is a limitation we return to later.

4.7 PCA: Can We Simplify the Picture?

With 6 features in play, we asked: can we reduce this complexity to just a few meaningful dimensions? Principal Component Analysis (PCA) provided the answer.

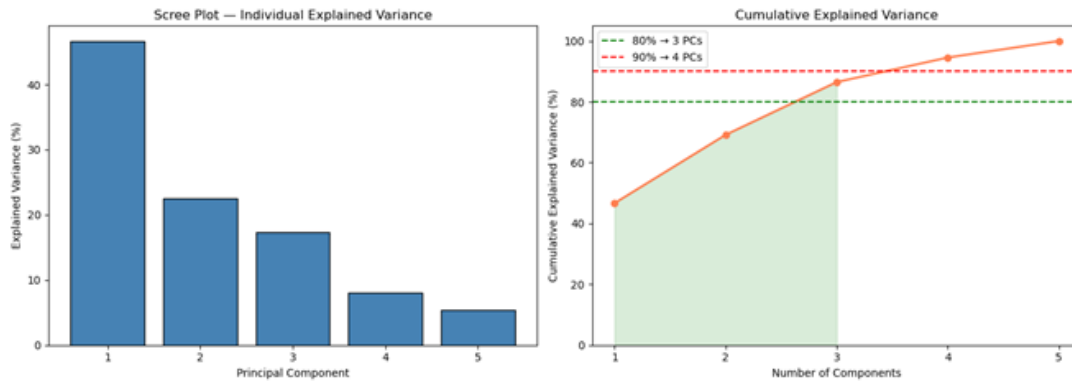


Fig 14: Scree and cumulative variance: no single component dominates. 3–5 are needed for 80 % coverage.

Component	Key Drivers	Interpretation
PC1	Square feet, bedrooms, bathrooms	"Apartment Size"
PC2	Latitude, longitude	"Geographic Position"

The biplot, coloured by price quartile, brings the story full circle:



Fig 15: PCA biplot: expensive apartments (red, Q4) cluster toward large PC1 (big) and specific geographic zones on PC2.

PCA confirms what every individual analysis has been pointing to: **size and location are the two dominant latent forces shaping rent**. Everything else adds much smaller, independent contributions.

5. Feature Engineering

Before building predictive models, we engineered features to capture information buried in raw text and categories:

- **15 amenity binary flags** – extracted via regex from the free-text **amenities** column.
- **pets_allowed_binary** – 1 if any pet policy exists, 0 otherwise.
- **is_luxury** – 1 if the title contains "luxury", "premium", "renovated", etc.
- **state_avg_price** and **state_median_price** target-encoded from training data only (log-scale) to capture regional rent levels **without data leakage**.

Before modelling, we checked for multicollinearity: VIF and Correlation Matrix:

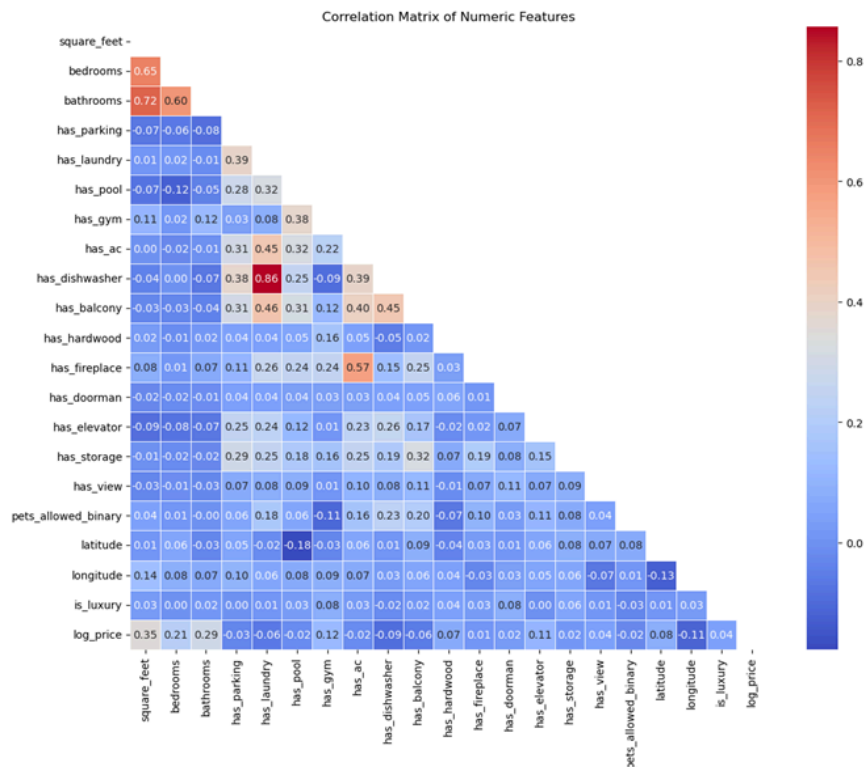


Fig 16: Correlation matrix: no severe multicollinearity between the 22 modelling features.

6. Predicting Rent

6.1 OLS Regression – Full Model (22 features, Log_Price)

Our baseline: a simple linear regression using all 22 features with standardised coefficients. **Test $R^2 = 0.524$** . The train-test gap is tiny (**+0.009**). No overfitting here, but the model leaves nearly half the rent variation unexplained.

The strongest coefficients confirm our EDA story: **state average rent** ($\beta = 0.318$) dominates, followed by **square feet** ($\beta = 0.130$) and **elevator** ($\beta = 0.043$, a proxy for urban high-rises). Most amenity flags – pool, fireplace, dishwasher – had insignificant p-values, confirming our suspicion that amenities are noisy once location and size are controlled.

6.2 OLS Reduced – Backward Elimination (10 features)

Can we do just as well with fewer features? Using backward elimination (removing the least significant predictor iteratively until ≤ 10 features with all $p < 0.05$), we trimmed 12 features.

Test $R^2 = 0.524$. Identical to the full model. The AIC is slightly higher (651.5 vs. 640.1), but the model is far more parsimonious.

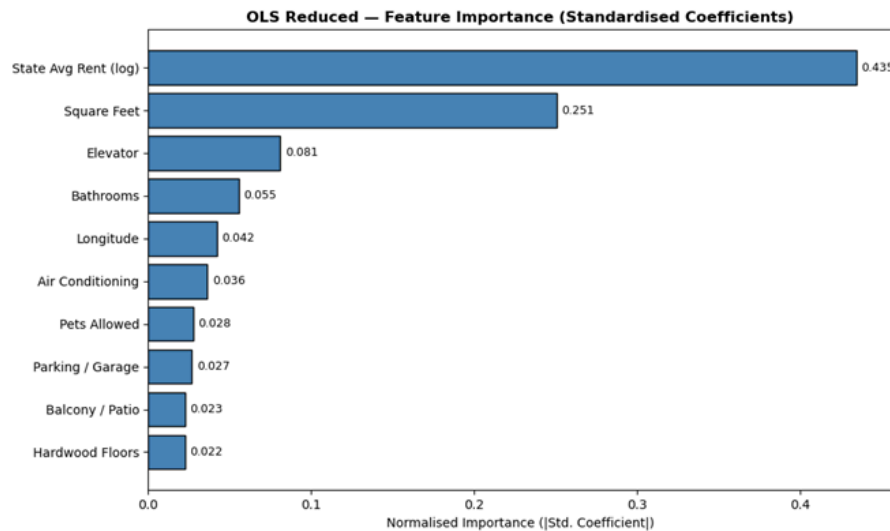


Fig 17: OLS Reduced: feature importance by normalised standardised coefficient. State average rent and square feet tower over everything else.

6.3 Random Forest (GridSearchCV)

Linear models cannot capture interaction effects. Maybe an elevator matters more in Manhattan than in rural Texas. We turned to Random Forest, tuning over 24 hyperparameter combinations with 5-fold cross-validation. Best configuration: 300 trees, max depth 16, min samples split 5, min samples leaf 4.

Test $R^2 = 0.69$. A major leap from OLS. The non-linear model captures patterns that a straight line cannot. However, the train R^2 of **0.849** reveals an R^2 gap of **0.156**, signalling moderate overfitting.

Top features: state_avg_price (**33.5 %**), square_feet (**23.5 %**), state_median_price (**11.5 %**), latitude (**11.0 %**), longitude (**9.8 %**).

6.4 CatBoost - Gradient Boosting

Our final and most powerful model: CatBoost, a gradient-boosting algorithm that builds trees sequentially, each correcting the errors of the last. Trained with **2 000** iterations, learning rate **0.05**, depth 7, L2 regularisation 5, and early stopping (best iteration: **1 997**).

Test $R^2 = 0.723$. The best performance across the board. CatBoost distributes importance more evenly than Random Forest: latitude (**18.7 %**), square_feet (**18.1 %**), longitude (**16.2 %**), state_avg_price (**13.5 %**), state_median_price (**12.1 %**).

6.5 Models Performance Comparison

Model	Test R^2	RMSE (\$)	MAE (\$)	MAPE (%)	Train R^2	R^2 Gap	Features
CatBoost	0.7225	261	185	15.3	0.857	0.135	22
Random Forest	0.6928	274	193	16.2	0.849	0.156	22
OLS Reduced	0.5243	341	252	21.0	0.530	0.006	10
OLS Full	0.5240	342	252	21.0	0.533	0.009	22

7. Methodological Limitations

No analysis is without blind spots. Ours has several:

- **A 3-month snapshot, not a longitudinal study.** The data covers Sep–Dec 2019. We cannot assess seasonal patterns, year-over-year trends, or the impact of events like COVID-19. Our conclusions are time-bound.
- **Geographic sampling bias.** The 10 000 listings are not a representative sample of the US market. Some states and cities are over-represented, biasing geographic conclusions.
- **Missing variables.** Building age, interior quality, transit proximity, neighbourhood safety, and lease terms are absent – likely explaining the ~28 % variance our best model cannot capture.
- **Amenity parsing via regex.** Misspellings, synonyms, or unusual phrasing may cause amenities to be missed or miscategorised.
- **Tree model overfitting.** R^2 gaps of 0.13–0.16 suggest the tree models may not generalise fully to data from different periods or regions.
- **Target encoding noise.** States with few listings produce noisy encoded values that may not generalise beyond the training set.

8. Conclusion

We began with a simple question: **what determines how much you pay for an apartment?** After the analysis the answer is clear:

1. **Location is king.** Where you live – your state, your city, your coordinates on a map – is the single most powerful predictor of rent.
2. **Size matters, with diminishing returns.** Square footage is the second most important factor. But the LOWESS analysis revealed a sub-linear relationship: you get a better deal on price-per-foot as apartments get larger, with the sweet spot at 3 bedrooms.
3. **Most amenities are marketing, not value.** Of 15 amenity features, only elevators, parking, and hardwood floors showed consistent significance. Pool, gym, fireplace – features that sound premium – add nothing to the price once location and size are accounted for.
4. **Non-linear models see what OLS cannot.** CatBoost outperforms OLS by a wide margin, proving that rent pricing involves complex interactions.
5. **~28 % remains a mystery.** Even our best model cannot explain about a quarter of rent variation, likely driven by factors invisible in this dataset: building age, interior finish, neighbourhood reputation, and lease terms.

9. AI Disclaimer

Tools used: GitHub Copilot, Gemini and ChatGPT

Declaration: AI tools were used to help improve the structure, create visualizations, and assist with coding. The main analysis, interpretations, and final results are the author's own original work.

10. Haiku

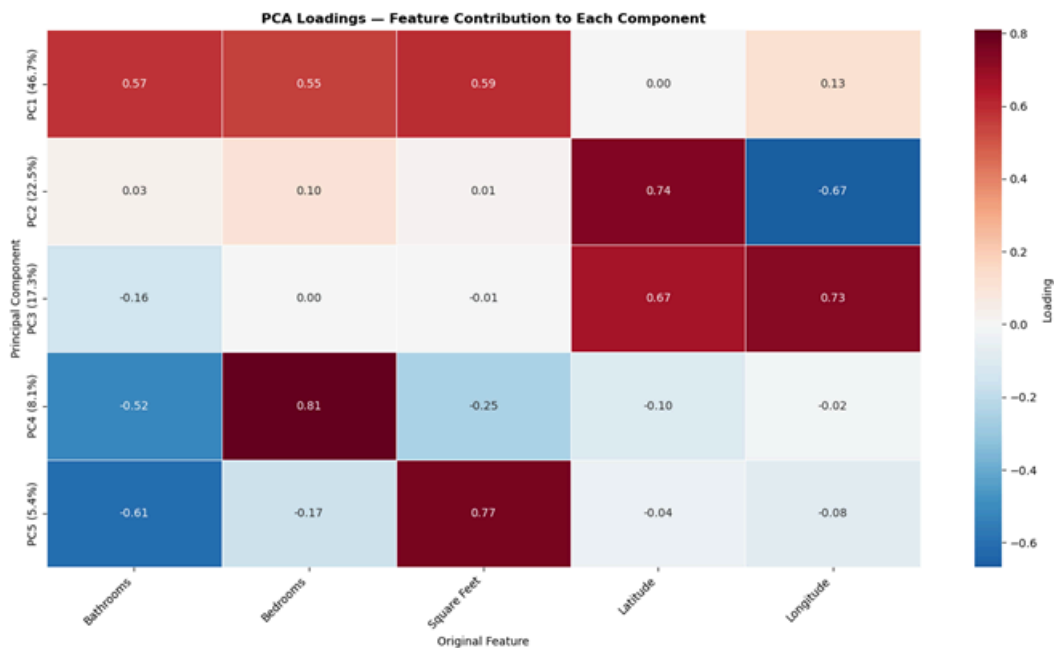
Rent drifts like the wind

Square feet, state lines shape the price.

Location wins all.

11. Appendix

1. **Tools:** Python 3.13, pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy, statsmodels, pingouin, scikit-learn, CatBoost. Analysis in Jupyter Notebook (VS Code).
2. **Reproducibility:** All code is in [ASDA_Project.ipynb](#). The dataset loads from HuggingFace. Random seed = 42 for all stochastic operations.
3. **PCA loading**

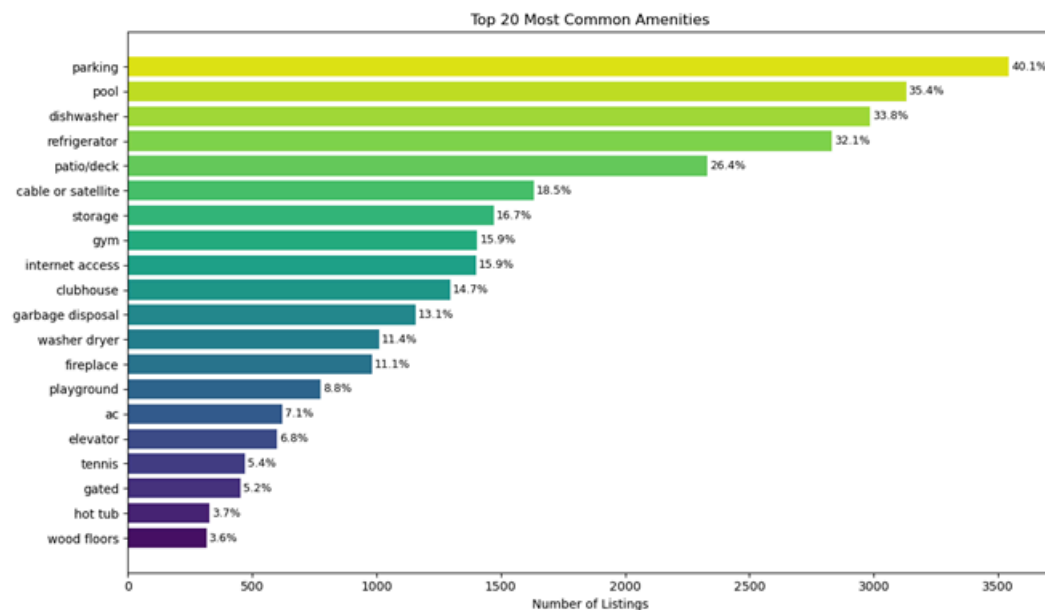


PCA loadings: PC1 is "apartment size" (sq ft, beds, baths); PC2 is "geographic position" (lat, long).

4. Do Amenities Matter?

Apartment listings often advertise amenities – parking, pool, gym, hardwood floors. But do these actually *move the price needle*, or are they just marketing?

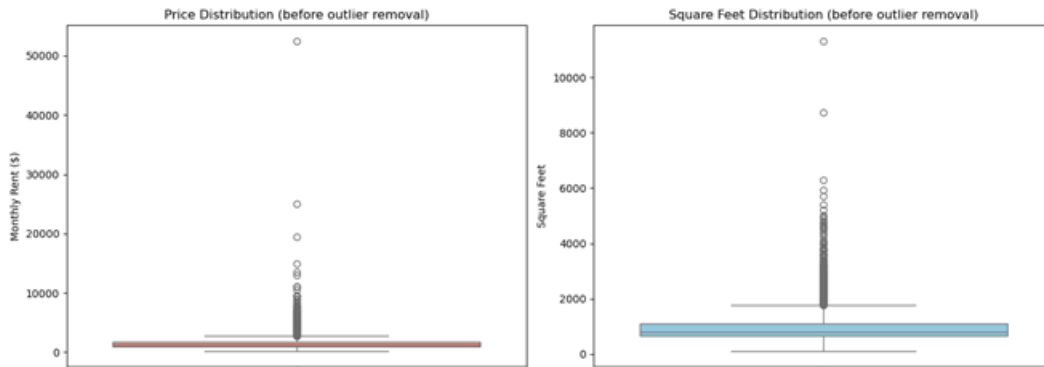
We parsed the free-text amenity field into 15 binary flags using regex pattern matching:



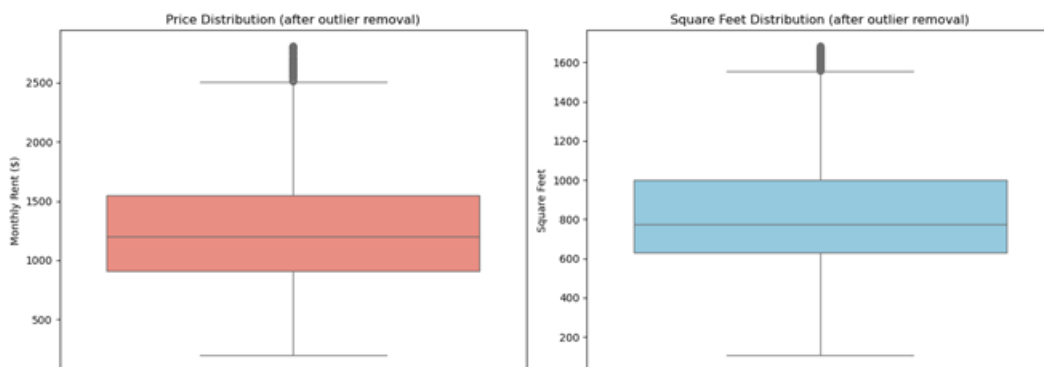
Top 20 amenities: laundry (41 %) and parking (40 %) are the most common; doorman (0.3 %) the rarest.

The question of whether amenities actually predict price is one we answer definitively in the modelling section. Spoiler: most of them are noise once you control for where the apartment is.

5. Outlier Removal for price and square feet:



Before outlier removal: extreme values stretch the distributions beyond any reasonable range.



After outlier removal: the data is now compact and ready for honest analysis.

6. Why CatBoost?

Handles categorical features natively – Unlike Random Forest or XGBoost, CatBoost can process categorical variables (like state) directly without manual one-hot encoding, using an efficient ordered target encoding internally.

Strong performance on tabular data – CatBoost is a gradient-boosting algorithm specifically optimized for structured/tabular datasets, which is exactly what the apartment rental data is.

Reduces overfitting – It uses ordered boosting (a permutation-driven approach) that reduces prediction shift and overfitting compared to traditional gradient boosting, which is important with only ~10K rows.

Minimal hyperparameter tuning needed – CatBoost works well out of the box with sensible defaults, making it a practical choice.

Benchmark against simpler models – In the project it served as the most advanced model in a progression (OLS -> Random Forest -> CatBoost), showing how much predictive power is gained by moving from linear to ensemble to gradient-boosted methods.

Resources: <https://arxiv.org/abs/1706.09516>, <https://arxiv.org/abs/2408.14817>